



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No - 10	Team No - 14	Project Title- Vehicle Service Booking & Tracking System
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1. Abstract

The **Vehicle Service Booking & Tracking System** is a backend-based application designed to simplify and manage the vehicle servicing process through centralized service booking and tracking. The system allows customers to register vehicles, select preferred service centers, check available service slots, book appointments, and track service progress. The backend manages bookings, job cards, technician assignments, vehicle inspections, spare-parts inventory, additional repair approvals, invoices, payments, and service history. It will be developed using **Node.js and Express.js for backend development and MySQL for database management**, with REST APIs connecting different application modules. Role-based access will be provided for customers, technicians, service staff, and administrators. Additional features such as location-based service-center selection, pickup and drop management, notifications, service reminders, customer verification, and service analytics will improve efficiency, transparency, and communication while maintaining organized and secure service records.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090008	B. Sathish Kumar	
25200900139	A. Shashank Reddy	
2520090071	D. Vignesh Kumar	

3. Frameworks / Platforms / Software Used

- ✓ **Node.js** – Backend runtime environment
- ✓ **Express.js** – Framework for developing REST APIs
- ✓ **MySQL** – Relational database management system
- ✓ **Sequelize ORM** – Database interaction and model management
- ✓ **JWT (JSON Web Token)** – User authentication and role-based authorization
- ✓ **bcrypt** – Secure password hashing

- ✓ **Postman** – API development and testing
- ✓ **VS Code** – Development environment
- ✓ **Git & GitHub** – Version control and project management
- ✓ **Razorpay API** – Online payment processing
- ✓ **Leaflet + OpenStreetMap** – Service-center location and map functionality
- ✓ **Multer** – Handling vehicle/service-related image uploads
- ✓ **Nodemailer** – Email notifications and service updates

Guide Name and Signature (remarks if any) _____